

Brian Borbath

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SUMMARY

Electrical Engineering freshman at the University of Florida passionate in semiconductor devices and hardware engineering with hands on experience in RF, embedded systems experimentation, and python programming.

EDUCATION

University of Florida

Expected May 2030

B.S. Electrical Engineering • Gainesville, FL

Relevant Coursework: Calculus III, C++ for Engineers, Applied Chemistry

TECHNICAL PROJECTS

Raspberry Pi Thermal Efficiency Research

- Designed and conducted an experimental investigation of how a 3D-printed case could improve the thermal efficiency of a Raspberry Pi running AI workloads.
- Developed Python-based data collection workflows to record performance and automatically conduct statistical analysis.

VHF Antenna Design/Testing

- Designed VHF antennas for amateur radio operation using wavelength-based element calculations.

Home Server & Network Infrastructure

- Configured Linux/Windows servers, VPN routing, DNS, reverse proxies, and TLS services.

Programmable Semiconductor Curve Tracer (In Progress)

- Design a semiconductor curve tracer to analyze the I-V characteristics of diodes and BJTs.
- Integrate an ESP32 microcontroller to automate data collection to send to a PC and characterize using Python.

EXPERIENCE

Gator Amateur Radio Club

August 2026 – Current

Secretary • UF

- Maintain meeting records, assist with club communications, and support coordination of events while managing the club website.

The Forge

June 2025 – August 2025

Lab Assistant • Patrick Space Force Base

- Contributed to innovation efforts by supporting real lab activities focused on reducing operational costs of the US Space Force with 3D printers and hardware scanners.

Civil Air Patrol, USAF Auxiliary

September 2022 – June 2026

Deputy Commander • Satellite Beach, FL

- Led and mentored 75+ cadets while supervising staff and maintaining squadron chain-of-command.
- Planned weekly meetings, training activities, and events in coordination with adult leadership.
- Dedicated 4+ hours a week to emergency services, aerospace education, and community service.

TECHNICAL SKILLS

Programming: Python, Java, C++

Tools/Platforms: Git, Linux, VS Code, Gradle, Raspberry Pi, MS Office

Engineering: RF antenna design/testing, soldering, 3D printing, electronics prototyping